



Summary of The Black Swan

Original book by Nassim Nicholas Taleb

Black Swans are extremely unpredictable events that have massive impacts on human society. These include positive Black Swans, like the invention of the Internet and the discovery of antibiotics, as well as negative Black Swans, like the 2008 recession.

Black Swans compel people to explain *why* they happened—to show, after the fact, that they were indeed predictable. Taleb's thesis, however, is that Black Swans, by their very nature, are always unpredictable. Furthermore, because of our cognitive biases, we're more vulnerable than ever to misunderstanding Black Swans and their impact.

Learn what Black Swans are, why we're susceptible to them, and how best to prepare yourself for the unpredictable.

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1-Page Summary

The Black Swan is the second book in former options trader Nassim Nicholas Taleb's five-volume series on uncertainty. This book analyzes so-called "Black Swans"—extremely unpredictable events that have massive impacts on human society.

The Black Swan is named after a classic error of induction wherein an observer assumes that because all the swans he's seen are white, all swans *must* be white. Black Swans have three salient features:

- They are rare (statistical outliers);
- They are disproportionately impactful; and, because of that outsize impact,
- They compel human beings to explain *why* they happened—to show, after the fact, that they were indeed predictable.

Taleb's thesis, however, is that **Black Swans, by their very nature, are *always* unpredictable**—they are the "unknown unknowns" for which even our most comprehensive models can't account. The fall of the Berlin Wall, the 1987 stock market crash, the creation of the Internet, 9/11, the 2008 financial crisis—all are Black Swans.

Once Taleb introduces the concept of the Black Swan, he delves into human society and psychology, analyzing why modern civilization invites wild randomness and why humans can neither accept nor control that randomness.

Extremistan vs. Mediocristan

To explain how and why Black Swans occur, Taleb coins two categories to describe the measurable facets of existence: Extremistan and Mediocristan.

In Mediocristan, randomness is highly constrained, and deviations from the average are minor.

Physical characteristics such as height and weight are from Mediocristan: They have upper and lower bounds, their distribution is a bell curve, and even the tallest or lightest human being isn't much taller or lighter than the average. *In Mediocristan, prediction is possible.*

In Extremistan, however, randomness is wild, and deviations from the average can be, well, extreme. Most social, man-made aspects of human society—the economy, the stock market, politics—hail from Extremistan: They have no known upper or lower bounds, their behavior can't be graphed on a bell curve, and individual events or phenomena—i.e., Black Swans—can have exponential impacts on averages.

Imagine you put ten people in a room. Even if one of those people is Shaquille O'Neal, the average height in the room is likely to be pretty close to the human average (Mediocristan). If one of those people is Jeff Bezos, however, suddenly the wealth average changes drastically (Extremistan).

The Unreliability of "Experts"

Taleb has very little patience for "experts"—academics, thought leaders, corporate executives, politicians, and the like. Throughout the book, Taleb illustrates how and why "experts" are almost always wrong and have little more ability to predict the future than the average person.

There are two reasons “experts” make bad predictions:

1) Human Nature

Because of various habits innate to our species—our penchant for telling stories, our belief in cause and effect, our tendency to “cluster” around specific ideas (confirmation bias) and “tunnel” into specific disciplines or methods (specialization)—we tend to miss or minimize randomness’s effect on our lives. Experts are no less guilty of this blindspot than your average person.

2) Flawed Methods

Because experts both (1) “tunnel” into the norms of their particular discipline and (2) base their predictive models exclusively on past events, their predictions are inevitably susceptible to the extremely random and unforeseen.

Consider, for example, a financial analyst predicting the price of a barrel of oil in ten years. This analyst may build a model using the gold standards of her field: past and current oil prices, car manufacturers’ projections, projected oil-field yields, and a host of other factors, computed using the techniques of regression analysis. The problem is that this model is innately narrow. It can’t account for the truly random—a natural disaster that disrupts a key producer, or a war that increases demand exponentially.

Taleb draws a key distinction between experts in Extremistan disciplines (economics, finance, politics, history) and Mediocristan disciplines (medicine, physical sciences). Experts like biologists and astrophysicists are able to predict events with fair accuracy; experts like economists and financial planners are not.

Difficulties of Prediction

The central problem with experts is their uncritical belief in the possibility of prediction, despite the mountain of evidence that indicates prediction is a fool’s errand. Some key illustrations of the futility of prediction include:

Discoveries

Most groundbreaking discoveries occur by happenstance—luck—rather than careful and painstaking work. The quintessential example is the discovery of penicillin. Discoverer Alexander Fleming wasn’t researching antibiotics; rather, he was studying the properties of a particular bacterium. He left a stack of cultures lying out in his laboratory while he went on vacation, and when he returned he found that a bacteria-killing mold had formed on one of the cultures. *Voilà*—the world’s first antibiotic.

Dynamical Systems

A dynamical system is one in which an array of inputs affect each other. Whereas prediction in a system that contains, say, two inputs, is a simple affair—one need only account for the qualities and behavior of those two inputs—prediction in a system that contains, say, five hundred billion inputs is effectively impossible.

The most famous illustration of a dynamical system’s properties is the “butterfly effect.” This idea was proposed by an MIT meteorologist, who discovered that an infinitesimal change in input parameters can drastically change weather models. The “butterfly effect” describes the possibility that the flutter of a butterfly’s wings can, a few weeks later and many miles distant, cause a tornado.

Predicting the Past

The past itself is as unknowable as the future. Because of how complex the world is and how a single event could be influenced by any number of tiny causes, we cannot reverse engineer causes for events.

An example should help illustrate. Think of an ice cube sitting on a table. Imagine the shape of the puddle that ice cube will make as it melts.

Now think of a puddle on the table and try to imagine how that puddle got there.

When historians propose causes for certain historical events, they're looking at puddles and imagining ice cubes (or a spilled glass of water, or some other cause). The problem is that the sheer number of possible causes for a puddle—or a historical event—render any ascription of cause suspect.

If You Can't Predict, How Do You Deal with Uncertainty?

Although Taleb is far more concerned with explaining why prediction is impossible than he is with proposing alternatives or solutions, he does offer some strategies for dealing with radical uncertainty.

1) Don't Sweat the Small Predictions

When it comes to low-stakes, everyday predictions—about the weather, say, or the outcome of a baseball game—there's no harm in indulging our natural penchant for prediction: If we're wrong, the repercussions are minimal. It's when we make large-scale predictions and incur real risk on their basis that we get into trouble.

2) Maximize Possibilities for Positive Black Swans

Although the most memorable Black Swans are typically the negatively disruptive ones, **Black Swans can also be serendipitous**. (Shortform note: Love at first sight is an example of a serendipitous Black Swan.)

Two strategies for opening ourselves up to positive Black Swans are (1) sociability and (2) proactiveness when presented with an opportunity. Sociability puts us in the company of others who may be in a position to help us—we never know where a casual conversation might lead. And proactiveness—for example, taking up a successful acquaintance on an invitation to have coffee—ensures we'll never miss our lucky break.

3) Adopt the "Barbell Strategy"

When Taleb was a trader, he pursued an idiosyncratic investment strategy to inoculate himself against a financial Black Swan. He devoted 85%–90% of his portfolio to extremely safe instruments (Treasury bills, for example) and made extremely risky bets—in venture-capital portfolios, for example—with the remaining 10%–15%. (Another variation on the strategy is to have a highly speculative portfolio but to insure yourself against losses greater than 15%.) The high-risk portion of Taleb's portfolio was highly diversified: **He wanted to place as many small bets as possible to increase the odds of a Black Swan paying off in his favor.**

The "barbell strategy" is designed to minimize the pain of a negative Black Swan while, potentially, reaping a positive Black Swan's benefits. If the market collapses, a person pursuing this strategy isn't hurt beneath the "floor" of the safe investments (say, 85%), but if the market explodes, he has a chance to capitalize by virtue of the speculative bets.

4) Distinguish Between Positive Contingencies and Negative Ones

Different areas of society have different exposure to Black Swans, both positive and negative. For example, scientific research and moviemaking are “positive Black Swan areas”—catastrophes are rare, and there is always the possibility of smashing success. The stock market or catastrophe insurance, meanwhile, are “negative Black Swan areas”—upsides are relatively modest compared to the possibility of financial ruin.

Suffice it to say, **we should take more risks in a positive Black Swan area than in a negative Black Swan one.**

5) Prepare, Don't Predict

Because Black Swans are, by definition, unpredictable, **we're better off *preparing* for the widest range of contingencies than *predicting* specific events.**

That's because, though Black Swans themselves can never be predicted, their *effects* can be. For example, no one can predict when an earthquake will strike, but one can know what its effects will be and prepare adequately to handle them.

The same goes for an economic recession. No one can predict precisely when one will occur, but, using the “barbell strategy” or some other means of mitigating risk, we can at least be prepared for one.

Shortform Introduction

In his April 2007 review of *The Black Swan* in the *New York Times*, Gregg Easterbrook, riffing on the author's skepticism about forecasts of any kind, noted, "At the beginning of 2006, the *Wall Street Journal* forecast a bad year for stocks; the Dow Jones Industrial Average rose 16% that year. (Disturbingly, the *Journal* has forecast a good year for 2007.)" Mere months later, the world economy would be in a tailspin—and Nassim Nicholas Taleb, who in *The Black Swan* warned that the global financial system was vulnerable to collapse, would be treated as a seer.

The Black Swan covers a broad range of topics and is organized in a somewhat unbalanced way. The first two parts of the book contain its core ideas, moving from (1) the ways we domesticate randomness to (2) the reasons why prediction is impossible to (3) humans' best options when it comes to uncertainty. (Chapters 1–7 of the summary correspond to these topics.) The third part of the book, meanwhile, is framed as an add-on to the first two parts for the more technically minded reader—it grounds many of the claims Taleb makes in the second part with examples from history and the social sciences. (This material can be found in our summary's "Appendix.") The fourth and final part, entitled "The End," is very brief and functions more or less as a farewell to the reader.

Taleb's knowledge is encyclopedic. Over the course of the text's four parts, he discusses mathematics, economics, philosophy, statistics, psychology, political science, history, physics, and literature, among other topics, all with an eye to exposing our assumptions about the nature of randomness and our abilities to account for it.

This summary covers all of Taleb's major claims as well as many of his real-world examples, but it doesn't attempt to recreate his energetic prose or his sense of humor. For those readers who want to experience Taleb's unique writing voice and personality, we urge you to turn to the book itself.

Chapter 1: What Is a Black Swan?

For millennia, it was universally accepted that all swans were white. In fact, this truth was so incontrovertible that logicians would often use it to illustrate the process of deductive reasoning. That classic deduction went like this:

1. All swans are white
2. The bird is a swan
3. The bird is white

But in 1697, Willem de Vlamingh, a Dutch explorer, discovered black swans while on a rescue mission in Australia—and, in an instant, a universal, incontrovertible truth was shown to be anything but.

After Vlamingh's discovery, philosophers used the term “black swan” to describe a seeming logical impossibility that could very well end up being possible.

Taleb, however, offers a new spin on the term. **He uses it to describe specific historical events with specific impacts.** These events have three salient features:

- They are “outliers” (that is, they are *statistically* insignificant);
- They have profound real-world impacts; and
- Despite (or perhaps because of) their extreme unpredictability, they compel human beings to account for them—to explain *after the fact* that they were in fact predictable.

Some examples of Black Swan events include World Wars I and II, the fall of the Berlin Wall, 9/11, the rise of the Internet, the stock-market crash of 1987, and the 2008 financial crisis.

Taleb's thesis is that Black Swans, far from being insignificant or unworthy of systematic study, comprise the *most significant* phenomena in human history. We *should* study them, even if we can't predict them. Thus, counter-intuitively, we would be better served by concentrating our intellectual energies on what we *don't*—nay, *can't*—know, rather than on what we *do* and *can* know.

Taleb also claims, also counter-intuitively, that **the more our knowledge advances, the more likely we are to be blindsided by a Black Swan.** This is because our knowledge is forever becoming more precise and specific and less capable of recognizing generality—for example, the general tendency for earth-shattering events to be completely unforeseen (which, of course, is *why* they're earth-shattering).

Platonicity and the Platonic Fold

Like Plato, with his abstract and ideal “forms,” human beings in general tend to favor neat, “pure” concepts that are universally consistent. These concepts—mathematical rules, notions of historical progress, economic laws—allow us to form models of the world so that predictions are much easier to make.

The problem with these models is that they lead us to “mistake the map for the territory”—that is, we are fooled into thinking the models *are* reality, rather than a very particular representation of reality that excludes outliers (i.e., Black Swans).

Taleb calls this natural human tendency to box in reality *Platonicity* and he holds it responsible for our dangerous confidence in our own knowledge. **We become so enamored of our elegant, self-consistent**

models that we are unable to see beyond them.

It's where our models cease to be useful that Black Swans occur—in the *Platonic fold* between our predictive models and unpredictable reality.

The Origins of Taleb's Black Swan Obsession

Taleb's first encounter with a Black Swan took place in his home country of Lebanon.

For centuries, the region around Mount Lebanon was known for its cosmopolitanism—located on the eastern shore of the Mediterranean, at the crossroads of Europe and the Near East, it was home to a vibrant mercantile population that included Christians and Muslims of various sects, Jews, and Druze.

This tolerant and multicultural paradise was still in existence in Taleb's youth. He recalls the motley crew of international playboys, spies, writers, and merchants that frequented the country.

Taleb was part of the country's aristocracy. Both of his grandfathers were educated in France, and one of them was serving as minister of the interior when a teenaged Taleb was jailed for participating in a political rally that turned violent. The government was scared enough of the unrest to grant all the arrested protesters amnesty.

Shortly after the rally came the Black Swan that launched Taleb's obsession: A civil war between Lebanon's Muslims and Christians that shattered the country's millennia-long ethnic peace and reduced Beirut, Lebanon's capital and the "Paris of the Middle East," to rubble.

What this Black Swan brought home for Taleb was the blunt unknowability of history. The war wasn't—couldn't have been—predicted; it could only be explained after the fact. He realized that human beings suffer from a "triplet of opacity" when it comes to our encounters with history:

1) The Illusion of Understanding

We all tend to *think* we have a grasp of what's going on in the world when, in fact, the world is far more complex than we know.

For example, all the adults around Taleb predicted the civil crisis would last a matter of days (it ended up lasting around 17 years). Despite the fact that events kept contradicting people's forecasts, people acted each day as though nothing exceptional had occurred.

2) The Retrospective Distortion

History always appears predictable (or, at least, explainable) in our retrospective accounts of events.

In the case of the Lebanese Civil War, **the adults whose forecasts were continually proved incorrect were always able to explain the surprising events *after the fact*.** In other words, the events always *were indeed* predictable, but one could only predict them *after they'd already happened*.

3) The Overvaluation of Facts and the Flaw of Expertise

We accumulate information and listen to expert analysis of that information, but these elements never measure up to real events.

Taleb cites the example of his grandfather, who eventually rose to deputy prime minister of Lebanon. Although his grandfather was an educated man with years of experience in politics, his forecasts were

proven wrong as routinely as those of his uneducated driver. **Neither knew more than the other about the twists and turns of the war.**

Newspapers, too, did nothing to help the Lebanese understand the war. They communicated information, but they didn't make anyone's predictions any more accurate. The reporters of the war also tended to "cluster"—emphasizing the same details and using the same categories as each other. Clustering reduces the complexity of the world—and leaves us vulnerable to Black Swans.

(When Taleb was attending Wharton, he learned about "efficient markets"—the idea that trading securities can't produce profits because all relevant information is publicly available and thus already built into prices. Upon absorbing this idea, Taleb completely ceased to read newspapers or watch television.)

In the 1980s, as the war continued to rage, Taleb found himself in business school at the University of Pennsylvania's Wharton School. There, he discovered that the triplet of opacity didn't just affect his countrymen—it affected his professors and the powerful corporate executives who came to lecture as well. Taleb realized that *no one*—not even the "smartest" businesspeople in the world—was prepared for Black Swan events.

That realization became even more stark on October 19, 1987, when global stock markets crashed.

At the time, Taleb was working for Credit Suisse First Boston; his specialty, honed at Wharton, was "quantitative finance"—the application of complex mathematical models to markets to mitigate uncertainty. Taleb, however, used his knowledge as a "quant" not to apply those mathematical models directly, but to suss out *where those models failed* and invest accordingly. In other words, whereas everyone else was flying blind and didn't know it, **Taleb was flying blind and *did* know it, and he placed bets that would pay off if the models were wrong.**

On the day of the crash, he was vindicated. So much so that his bonus was tantamount to "F*** you" money—that is, it was large enough that he no longer needed to work to earn money. He chose to stay in finance, but he customized his roles so that he could spend ample time in solitude and contemplation.

The focus of his sabbaticals? *Uncertainty*.

Chapter 2: Scalability | Mediocristan and Extremistan

One reason that Black Swans are so profoundly disruptive is that they occur in the “scalable” parts of our lives—where physical limits don’t apply and effects tend toward incredible extremes. When a particular thing—an income, an audience for a particular product—is “scalable,” it can grow exponentially without any additional expenditure of effort.

“Massage therapist,” for example, is a “nonscalable” profession. There is an upper limit on how many clients you can see—there’s only so much time in a day, and therapists’ bodies fatigue—and thus there’s only so much income you can expect from that profession.

“Quantitative trader,” however, is a “scalable” profession. It takes no additional energy or time to purchase 5,000 shares of a stock than 50, and your income isn’t limited by physical constraints.

Artists, too, are in a scalable profession (at least in the age of digital reproduction). For instance, a singer doesn’t need to perform her hit song each time someone wants to hear it. She performs it once for the record, and that performance can be disseminated widely.

The problem with scalability is that it creates vast inequalities. Let’s look at the singer example again:

- *Before the advent of recording technology*, a singer’s audience was limited to those for whom she could physically perform. That is, a singer in one town wasn’t likely to prevent the survival of a singer in another town; they might have differently sized audiences—based on the populations of their respective towns—and thus different incomes, **but those differences would be comparatively mild.**
- *After the advent of recording technology*, however, a small number of singers come to dominate the listening public. Now that we can pay pennies to stream Beyoncé any time we want, why spend the \$10 or \$20 to see a local singer we’ve never heard of? Suddenly, differences in audience and income become vast. **With scalability comes extremes.**

The Contrary Worlds of Mediocristan and Extremistan

“Mediocristan” is Taleb’s term for the facets of our experience that are nonscalable. For example, like the income of a massage therapist, human physical traits such as height and weight hail from Mediocristan—they have upper and lower bounds, and if you were to graph every human’s height and weight, you would produce a bell curve.

Mediocristan’s overriding law can be stated thus: **Given a large-enough sample size, no individual event will have a significant effect on the total.** That is, there will be outliers—extremely heavy or tall people—but those outliers (1) will not be *exponentially* larger or smaller than the average, and (2) will be rendered insignificant by the sheer number of average cases. Most physical phenomena—human footspeed, trees’ rate of growth—come from Mediocristan. (Shortform note: Taleb sometimes treats Mediocristan as a distinct place, other times as an adjective to describe certain kinds of phenomena.)

“Extremistan,” oppositely, describes those facets of our experience that *are eminently* scalable. **In Extremistan, inequalities are vast enough that one instance can profoundly affect the total.**

Most social (man-made) phenomena come from Extremistan. For example, wealth: It has no readily detectable upper limit; and if you were to include, say, Jeff Bezos, in any average of human wealth, you would produce a grossly distorted picture of how much money most people have.

Extremistan—Where Black Swans Fly

In the realm of Mediocristan, randomness is highly constrained (mild): There's only so much variation in the physical aspects of our world. **Thus, in Mediocristan, *Black Swans are (effectively) impossible*.**

In Extremistan, however, randomness is highly variable (wild): No matter how large your sample size for a given phenomenon, you can't produce a trustworthy average or aggregate picture because of the *variation* in that phenomenon. **In Extremistan, *Black Swans are frequent*.**

(Thanks to the work of French mathematician Benoît Mandelbrot, some Black Swans can become "Gray" Swans—rare and surprising events that *can* be imaginable, if not precisely predicted. See the Appendix for the theory behind Gray Swans.)

Key Qualities of Mediocristan	Key Qualities of Extremistan
Nonscalable	Scalable
Typical member is mediocre ("average," in the statistical sense)	Most members are dwarfs, a few are giants
Best-off are only marginally better than worst-off	Best-off are considerably better off than worst-off
Events are predictable from available information	Events are highly unpredictable from available information
Probability distribution is a bell curve	Probability distribution accords either with Mandelbrotian "Gray" Swans or is dominated by Black Swans

Chapter 3: Don't Be a Turkey | It Pays to Be a Skeptic

Picture a turkey cared for by humans. It has been fed every day for its entire life by the same humans, and so it has come to believe the world works in a certain, predictable, and advantageous way. And it does...until the day before Thanksgiving.

Made famous by British philosopher Bertrand Russell (though, in his telling, the unlucky bird was a chicken), this story illustrates the problem with *inductive reasoning* (the derivation of general rules from specific instances). **With certain phenomena—marketing strategy, stock prices, record sales—a pattern in the past is no guarantee of a pattern in the future.**

In Taleb's words, the turkey was a *sucker*—it had full faith that the events of the past accurately indicated the future. Instead, it was hit with a Black Swan, an event that completely upends the pattern of the past. (It's worth noting that the problem of inductive reasoning *is* the problem of Black Swans: Black Swans are possible because we lend too much weight to past experience.)

Another example of faulty inductive reasoning, this time from the world of finance, concerns the hedge fund Amaranth (ironically named after a flower that's "immortal"), which incurred one of the steepest losses in trading history: \$7 billion in less than a week. Just days before the company went into tailspin, Amaranth had reminded its investors that the firm employed twelve risk managers to keep losses to a minimum. The problem was that these risk managers—or suckers—based their models on the market's *past performance*.

In order not to be suckers, we must (1) cultivate an "**empirical skepticism**"—that is, a skepticism steeped in fact and observation—and (2) remain vigilant against the innately human tendencies that leave us vulnerable to Black Swans.

Traits of the Empirical (a-Platonic) Skeptic	Traits of the Platonifier
Respects those who say "I don't know"	Views those who say "I don't know" as ignorant
Thinks of Black Swans as a primary incidence of randomness	Thinks of minor deviations as the primary incidence of randomness
Minimizes theory	Praises theory
Assumes the world functions like Extremistan rather than Mediocristan	Assumes the world functions like Mediocristan rather than Extremistan
Prefers to be broadly right across a wide range of disciplines and situations	Prefers to be perfectly right in a narrow range of disciplines and situations

The Long History of Empirical Skepticism

The problem of induction illustrated by the turkey story has been noted by many well-known philosophers, including the great skeptic David Hume. But induction's shortcomings were noted even in antiquity.

Empirical Skeptic #1: Sextus Empiricus

Either a philosopher himself or simply a copyist of other thinkers, Sextus Empiricus resided in 2nd-century BC Alexandria. In addition to his philosophical pursuits, he practiced medicine, doing so according to empirical observation but without dogmatism (that is, without blind loyalty to a particular approach or method). In fact, Sextus Empiricus was devoutly antidogmatic: He eschewed theory of any kind—the title of his most famous book translates “Against the Professors”—and proceeded according to persistent trial and error, much like Taleb.

Empirical Skeptic #2: Al-Ghazali

An 11th-century Persian philosopher, Al-Ghazali too doubted the wisdom of the intellectual authorities of his time. (The title of his most famous text is *The Incoherence of the Philosophers*.) He expressed skepticism of “scientific” knowledge (as espoused by his rival, Averroës, who was himself influenced by Aristotle). Unfortunately Al-Ghazali’s ideas were coopted and exaggerated by later Sufi scholars, who argued that humans were better served by communing with God and leaving behind all earthly matters.

Empirical Skeptic #3: Pierre Bayle

A French skeptic of the 17th century, Bayle is best known for his *Historical and Critical Dictionary*, which critiques much of what passed for “truth” in his historical moment.

Empirical Skeptic #4: Pierre-Daniel Huet

A contemporary of Bayle, Huet, long before David Hume, proposed that for any event there could be an infinity of causes.

What Empirical Skeptics Resist

Empirical skeptics tend to resist five cognitive flaws that filter the truth and prevent the recognition of Black Swans.

Flaw #1: The Error of Confirmation

All too often we draw *universal* conclusions from a *particular* set of facts. For example, if we were presented with evidence that showed a turkey had been fed and housed for 1,000 straight days, we would likely predict the same for day 1,001 and for day 1,100.

Taleb calls this prediction the “round-trip fallacy.” **When we commit the round-trip fallacy, we assume that “no evidence of x”—where x is any event or phenomenon—is the same as “evidence of no x.”**

For example, in the turkey illustration, we might assume that “no evidence of the possibility of slaughter” equals “evidence of the impossibility of slaughter.” To take a medical example, if a cancer screening comes back negative, there is “no evidence of cancer,” not “evidence of no cancer” (because the scan isn’t perfect and could have missed something).

In addition to drawing broad conclusions from narrow observations, we also have a tendency to *select* evidence on the basis of preconceived frameworks, biases, or hypotheses. For example, a scientist conducting an experiment may, often unconsciously, discount evidence that disconfirms her hypothesis in favor of the evidence that confirms it. Taleb calls this habit “naive empiricism,” but it’s more commonly known as “confirmation bias.”

Taleb's solution to naive empiricism/confirmation bias is *negative empiricism*—the rigorous search for disconfirming, rather than corroborating, evidence. This technique was pioneered by a philosopher of science named Karl Popper, who called it “falsification.” The reason negative empiricism/falsification is so effective is that **we can be far more sure of wrong answers than right ones.**

Flaw #2: The Narrative Fallacy

Because humans are naturally inclined to stories, with distinct causes and effects, we are perennially in danger of committing the “narrative fallacy”—the ascription of meaning or cause to random events.

Our tendency to narrativize is part and parcel of our compulsion to *interpret*. Humans are evolutionarily conditioned—by the development of the left hemisphere of our brains—to reduce the complexity of the world's information (we'll discuss why in a moment); and the most efficient way of simplifying that complexity is through interpretation.

For example, read the following poem:

All work and no
no play makes Jack
a dull boy

Notice anything strange? (There's an extra “no” in the second line.) When experimental subjects reading a similar poem had the left-hemispheres of their brains impaired, they were better able to notice the extra “no.”

Neurotransmitters in the brain, too, encourage interpretation. When patients are administered dopamine supplements, they become more likely to see patterns where there are none.

Why are humans predisposed to interpretation? For a very practical reason: It makes information easier for our brains to store. Whereas retaining 100 randomly ordered numbers would be near impossible, retaining 100 numbers that were ordered according to a specific *rule* would be much easier. **When we interpret—or narrativize—we're attempting to impose our own organizing rule on the random facts of the world.**

So if we're biologically *and* psychologically conditioned to narrativize—and, thus, to remain blind to the totally random (i.e., a Black Swan)—what choice do we have? Taleb offers two suggestions:

Solution #1: Favor System 2 over System 1

Empirical psychologists like Nobel Laureate Daniel Kahneman and Amos Tversky have distinguished two types of thinking in human beings: “System 1” thinking, which is experiential, kneejerk, intuitive, and effortless; and “System 2” thinking, which is cogitative, slow, and effortful.

System 1 thinking is what kicks in when we immediately situate an event in a particular framework or pattern. For example, when we hear “no evidence of cancer” and think “evidence of no cancer,” we're receiving the information using our System 1 resources.

When faced with new information, we have to make a concerted effort to engage System 2 thinking—to pause and reason about the real significance of the information.

Solution #2: Favor Experiments Over Stories

Rather than trust in “expert” analyses or attempt to extrapolate cause-and-effect narratives from newspapers—materials that are typically based on inductive reasoning—**we should seek out empirical**

experimental evidence (for example, studies conducted by empirical psychologists like Kahneman and Tversky).

Flaw #3: The Disapproval of Society

We invite the disdain of our peers when we structure our lives around the pursuit of “good” Black Swans (for example, writing a novel in the hope it’ll be a bestseller) or the avoidance of bad ones (for example, adhering to a stock-trading strategy that shields us from a catastrophic event).

This is because humans are naturally inclined toward regular, smaller rewards rather than large windfalls. From the outside, the inventor who toils in his garage in pursuit of a world-changing innovation, or the stock trader who endures persistent losses to inoculate himself against a massive crash (which is what Taleb does), seems foolish. (In fact, it *is* foolish, at least in terms of financial rewards: Research shows that independent inventors earn at lower rates than venture capitalists.)

Nevertheless, inventors, artists, and iconoclastic stock traders continue to exist. They are able to endure the condescension of their peers and the perception of failure. They persist on *hope*.

These individuals, the hopeful, are what Taleb calls “reverse turkeys”—those who are prepared for Black Swan events, good or bad.

In his life as a trader, Taleb was a reverse turkey. He used a strategy called “bleed,” taking positions that would pay handsomely if there was a catastrophe but produced losses on a daily basis. He was, essentially, betting on lightning to strike. In 1987, it did.

Flaw #4: The Distortion of Silent Evidence

History—which Taleb defines as “any succession of events seen with the effect of posterity”—is inevitably, necessarily distorted. That is, no matter how “factual” or “objective,” the historical record is always a product of our tendency to narrate and thus always *biased*.

What the narratives are biased *against* is *randomness*—the inexplicability and importance of Black Swans.

Take most CEOs’ and entrepreneurs’ (auto)biographies. These books attempt to draw a causal link between the CEO/entrepreneur’s (a) character traits, education, and business acumen and (b) later success. The “silent evidence” (which Taleb also calls “the cemetery”) in these narratives **is that there are many more people with the same attributes as the triumphant CEOs/entrepreneurs who *failed***. The fact is, in business, like in so many other fields, **the deciding factor is nothing other than *luck* (i.e., randomness).**

Once we become attuned to the existence of “silent evidence”—which we can think of as the “flipside” or contrary to any story we’re told—we can see it everywhere.

For example, we base our knowledge of criminals and criminality on those that get caught (because the only criminals we know about are the ones that *get* caught). In other words, we have no real sense of how easy or hard it is to get away with a crime, because our notions about crime are formed solely by reports of *failed criminals*.

In the case of criminality, our ignorance of silent evidence actually serves a socially positive function because it makes getting away with a crime look hard and thus dissuades would-be criminals. But silent evidence can also encourage socially detrimental behaviors, such as risk-taking. This is because the successful risk-takers are the ones whose story is told; the failed risk-takers end up in the cemetery (in Taleb’s sense, but often literally as well).

Take, for example, Giacomo Casanova, the legendary ladies' man and adventurer. Casanova was blessed with infallible luck—somehow he always managed to escape whatever predicament he found himself in. He believed he had a “lucky star” that watched out for him.

But for every Casanova, there are dozens of wannabe adventurers and seducers whose risks don't pay off. We get a skewed sense of the benefits of risk-taking because the successful risk-takers are the ones with the means to tell their tales.

(Shortform note: Taleb doesn't acknowledge that stories of risks that fail spectacularly—from Bernie Madoff to Theranos to Lehman Brothers—*do* get told quite often.)

A more far-reaching variation on the concept of silent evidence is the “anthropic cosmological argument.” This argument, touted by physicists and philosophers alike, states that human existence cannot be a random occurrence because of the specificity and amount of factors that *provide* for that existence. In other words, the odds are so stacked against the fact of human existence, that the only possibility is that the world was created precisely to *allow* for our existence.

Again, what the anthropic cosmological argument suppresses is the evidence of all the other species that *didn't* thrive. We assume our beating the odds is the result of destiny, but it's really a matter of numbers: **If you consider the incredible number of species competing for survival, it stands to reason *one*—humans, in this case—was going to win the jackpot.**

Flaw #5: Our Tendency to “Tunnel”

A repercussion of the Distortion of Silent Evidence, “tunneling” describes the natural human tendency to favor *knowns* and *known unknowns* rather than *unknown unknowns*. In other words, **our understanding of uncertainty is based almost exclusively on what *has happened* in the past rather than what *could have happened*.**

The primary practitioners of tunneling are those Taleb calls “nerds”—academics, mathematicians, engineers, statisticians, and the like. Nerds are those who think entirely “inside the box”; they Platonify the world and can't perceive possibilities that lie outside their scientific models and academic training.

Nerds suffer from the “ludic fallacy.” (“Ludic” comes from the Latin word *ludus*, which means “game.”) That is, they treat uncertainty in real life like uncertainty in games of chance, for example roulette or blackjack. **The problem with this approach is that, unlike games of chance, *real life has no rules*.**

Nerds aren't the only ones guilty of the ludic fallacy, however; average people indulge it as well. For example, most people think casino games represent the height of risk and uncertainty. In truth, casino games hail from Mediocristan—there are clear and definite rules that govern play, and the odds of winning or losing are calculable. **Unlike real life, the amount of uncertainty in a casino game is highly constrained.**

The incredible unpredictability of real life can be illustrated by a look at casinos themselves, rather than their games. Casinos spend huge sums of money on sophisticated security systems—cameras, dealer training, alarms—in an effort to prevent cheating. These systems are designed to foil cheats who use methods that are either known or expected. But the casino Taleb studied suffered its worst losses not from cheating but from completely unexpected real-life events:

- an entertainer was attacked by the tiger with which he performed, resulting in a \$100 million loss due to cancelled shows;
- an employee, for years and for no good reason, neglected to file an essential tax form with the IRS,

resulting in a huge fine; and

- The casino owner's daughter was kidnapped, and to pay the ransom, the casino owner withdrew cash from the casino's fund.

Taleb's study of this particular casino yielded the insight that unexpected, **real-life events caused greater losses to the casino than cheating by a factor of almost 1,000 to 1**. Suffice it to say, to prepare ourselves for Black Swans, we must resist the ludic fallacy and think outside the rules of the games to which we're accustomed

Exercise: Responding to Randomness

Explore what it means to be an "empirical skeptic."

Write down something that happened to you recently, good or bad, that was out of the ordinary.

How did you respond to this event? Write down what you thought about it at the time.

Now, analyze your response to the event using Taleb's categories. Did you try to wrap a narrative around something that was truly random ("The Narrative Fallacy")? Did you treat what happened as a particular instance of a more general rule (Confirmation Bias, or "The Distortion of Silent Evidence")? Did you focus more on what happened than what *could have happened* (a consequence of "tunneling")? Explain why or why not.

Chapter 4: The Scandal of Prediction

With the rapid advance of technology—computer chips, cellular networks, the Internet—it stands to reason that our predictive capabilities too are advancing. But consider how few of these groundbreaking advances in technology were *themselves* predicted. For example, no one predicted the Internet, and it was more or less ignored when it was created.

(Shortform note: It's unclear how Taleb defines "predicted." Plenty of science-fiction writers and cultural commentators anticipated recent technologies like the Internet and augmented and virtual reality.)

It is an inconvenient truth that humans' predictive capabilities are extremely limited; we are continuously faced with catastrophic or revolutionary events that arrive completely unexpectedly and for which we have no plan. Yet, nevertheless, we maintain that the future is knowable and that we can adequately prepare for it. Taleb calls this tendency the *scandal of prediction*.

Epistemic Arrogance

The reason we overestimate our ability to predict is that we're overconfident in our knowledge.

A classic illustration of the fact comes from a study conducted by a pair of Harvard researchers. In the study, the researchers asked subjects to answer specific questions with numerical ranges. (A sample question might be, "How many redwoods are there in Redwood Park in California?" To which the subject would respond, "I'm 98% sure there are between x and y number of redwoods.") The researchers found that the subjects, though they were 98% sure of their answers, ended up being wrong 45% of the time! (Fun fact: The subjects of the study were Harvard MBAs.) In other words, the subjects picked overly narrow ranges because they *overestimated their own ability to estimate*. If they had picked wider ranges—and, in so doing, acknowledged their own lack of knowledge—they would have scored much better.

Taleb calls our overconfidence in our knowledge "epistemic arrogance." On the one hand, we overestimate what we know; on the other, we underestimate what we don't—*uncertainty*.

It's important to recognize that Taleb isn't talking about how much or how little we actually know, but rather the disparity between what we know and what we *think* we know. ***We're arrogant because we think we know more than we actually do.***

This arrogance leads us to draw a distinction between "guessing" and "predicting." Guessing is when we attempt to fill in a *nonrandom* variable based on incomplete information, whereas predicting is attempting to fill in a *random* variable based on incomplete information.

Say, for example, someone asks you to estimate how many natural lakes there are in Georgia. There's a right answer to the question—it's 0—but you don't know it, so your answer is a "guess."

But say that same someone asks you what the U.S. unemployment rate will be in a year. You might look at past figures, GDP growth, and other metrics to try and make a "prediction." **But the fact is, your answer will still be a "guess"**—there are just too many factors (unknown unknowns) to venture anything better than a guess.

The Curse of Information

It stands to reason that the greater our information is about a particular problem, the more likely we are to

come upon a solution. And the same goes, it would seem, for predictions: The more information we have to *make* a prediction, the more accurate our prediction will be.

But an array of studies shows that **an increase in information actually has negligible—and even negative—effects on our predictions.**

For example, the psychologist Paul Slovic conducted a study on oddsmakers at horse tracks. He had the oddsmakers pick the ten most important variables for making odds, then asked the oddsmakers to create odds for a series of races using only those variables.

In the second part of the experiment, Slovic gave the oddsmakers ten *more* variables and asked them to predict again. **The accuracy of their predictions was the same** (though their *confidence* in their predictions increased significantly).

The negative outcome of an increase in information is that we become increasingly sure of our predictions even as their accuracy remains constant.

Experts—The Worst Offenders

The most arrogant group in terms of their predictions—and least aware of their own ignorance—are so-called “experts.” These are the credentialed and/or laureled people whose opinions are granted weight by society.

Taleb divides this group by two. There are those who are arrogant but also display some degree of competence, and then there are those who are arrogant *and* incompetent.

1) Competent Arrogants

“Competent Arrogants” are those experts with actual predictive abilities and discernible skills. This group includes **astronomers, physicists, surgeons, and mathematicians** (when dealing exclusively with pure, rather than applied, mathematics).

2) Incompetent Arrogants

“Incompetent Arrogants” are “experts” whose predictive abilities and skills aren’t significantly greater than the average person’s. This group includes **stockbrokers, intelligence analysts, clinical psychologists, psychiatrists, economists, finance professors, and personal financial advisors.**

In numerous empirical studies, the forecasting ability of the “incompetent arrogants” has been shown to be almost nonexistent.

For example, over the course of several years, psychologist Philip Tetlock asked almost 300 specialists—political scientists, economists, journalists, and politicians—to offer predictions of world events (the timeframe was usually “within the next five years”). **He discovered that the experts’ predictions were barely more accurate than random selection and often worse than simple computer simulations.**

He also found that the more prominent a person was in his or her field, the worse were his or her predictions. The reason for this finding was that prominent people tend to become prominent based on their having *one big idea*. These experts marry themselves to their singular idea and neglect other possibilities—and thus, when randomness rears its head, they’re shown to have been woefully misguided.

Tetlock’s main interest, however, wasn’t the fact of experts’ poor forecasting abilities but rather those experts’ *lack of accountability* for being wrong. He found that experts (unconsciously) employ a number of

excuses to explain away their errors.

The “Different Game” Defense

Experts will claim that the event that disproved their prediction could have been predicted if the right data were available. For example, a specialist in the Soviet Union who failed to predict its collapse might say that, because the Soviet Union was so adept at hiding its economic data, he wasn’t able to make an accurate prediction.

The “Almost Right” Defense

Experts will claim that, if a minor variable or two had been different, they would have been proven correct. For example, a specialist in the Soviet Union who predicted that it would collapse twenty years after it actually did might say something like, “Well, I knew it *would* collapse, I just didn’t know exactly *when*.”

The “Outlier” Defense

Experts will claim that the event they failed to predict was a complete anomaly, a thousand-year flood—a Black Swan. For example, a specialist in the Soviet Union will claim its collapse *couldn’t* have been predicted, and so her inability to predict it doesn’t mean her predictive ability is suspect.

Problems with Projections

Like other predictions, projections—of incomes, costs, price fluctuations, construction time, and the like—are notoriously inaccurate.

This is because the authors of projections, like the authors of other kinds of predictions, “tunnel”—that is, they exclude from their calculations events external to whatever method they’re using.

Taleb cites the example of oil prices. In 1970, U.S. officials projected that the price of a barrel of oil would remain static or decline over the next ten years. In fact, due to crises like the Yom Kippur War and the Iranian Revolution, crude oil prices rose *tenfold* by 1980. (Shortform note: One can imagine those U.S. officials employing the “Different Game” and “Outlier” defenses to explain why their projection proved incorrect.)

A major problem with financial and governmental projections is that **they don’t incorporate a margin of error**. That is, the authors of these projections don’t take into account (nor do they publicize) how significantly their projections might be off-target.

The disregard for and omission of margins of error reveal three fallacies of projection:

1) The “Final Number” Fallacy

Because corporate projections omit a margin of error, we tend to fixate on the final number of the projection, taking it as gospel—when, in fact, it obscures a (wide) range of possibilities. For example, there’s a big difference between a projected ocean temperature rise of 1°C with a margin of error of 0.05°C and a projection of 1°C with a margin of error of 5°C.

2) The “Far Future” Fallacy

The further into the future one projects, the wider is that projection’s margin for error (because of the greater possibility for random occurrences), yet we treat these projections similarly to shorter-term

projections. Classic examples come from literature: George Orwell's *1984* (published in 1949), for example, was far off in terms of the state of the world in the mid-Eighties.

3) *The "Black Swan" Fallacy*

We underestimate the randomness of the variables used in the projection—that is, we fail to understand that any part of the method used to determine the projection is susceptible to Black Swans. For an example, see the oil-price-projection example just above.

Predictors = Liars

When leveling his critiques at financial, political, and security analysts—people who make their living from forecasting—Taleb often gets asked (snippily) to propose *his own* predictions. In these situations, **Taleb freely admits that he cannot forecast and that it would be irresponsible to attempt to.**

In fact, he goes a step further: He encourages those who forecast uncritically—the “incompetent arrogants”—to get new jobs. To him, bad forecasters are either fools or liars and do more damage to society than criminals.

Exercise: Learning the Limits of Prediction

Think like Taleb about prediction and the limits of your ability to predict things.

Write down a prediction you recently made, whether it was about a baseball game, the economy, an election, or other event.

Now explore *why* you made that particular prediction. What was your reasoning? What variables did you factor in?

Revisit your answer just above in light of Taleb's assertions. Did you rely on inductive reasoning? On the advice of an expert? On something you read or heard? Explain how you *developed* your reasoning.

In the space below, explore the possible gaps in your reasoning. What might you have left out? Was there

another side of the story ("silent evidence") you didn't consider?

Chapter 5: Why We Can't Know What We'll Know

Epistemic arrogance, the pretensions of “experts,” our ever-increasing access to information—all belie an incontrovertible fact: **In many, perhaps even most, areas of our lives, prediction is simply impossible.**

Take discoveries, for example. At any given moment, there are scores of scientists, scholars, researchers, and inventors around the world working diligently to better our lives and increase our knowledge. But what often goes unremarked is that **the discoveries with the profoundest impact on our lives are inadvertent—*random*—rather than the reward for careful and painstaking work.**

The discovery of penicillin is a case in point. Biologist Alexander Fleming left a stack of cultures sitting out in his laboratory while he went on vacation, and when he returned, a bacteria-killing mold had formed on one of the cultures. *Voila!*—the world's first antibiotic.

The same goes for the discovery of the cosmic microwave background, the omnipresent radiation in space that provides a key piece of evidence for the Big Bang. No researcher had any idea it existed until two radio astronomers noticed a hiss in their listening devices. How unexpected was their discovery? At first, the astronomers thought the hiss was caused by pigeon droppings on their antenna!

Because earth-shattering discoveries are unexpected or inadvertent, their importance, at least in the beginning, often goes unrecognized. When Darwin presented his findings at the Linnean Society, the 19th-century's preeminent institution of natural history, its president dismissed the theory as “no striking discovery.”

The Scientific Grounds for Uncertainty

The innovator of the scientific method of “falsification,” Karl Popper, also proposed an influential theory of history. The theory held that, because technological advance was “fundamentally unpredictable,” so too was the course of history.

Popper's theory is echoed by a key law in statistics called “the law of iterated expectations.” The law describes a situation **when to predict something means to *already know* that something.**

Think, for example, of the invention of the wheel. If a primitive human were to *predict* the invention of the wheel, that human would *already know enough* about the wheel to invent it him- or herself.

The same condition applies to *contemporary* predictions of discoveries. For example, some have predicted that carbon-capture technology will solve global warming. But to make that prediction, one has to know the specifications of that technology—that is, one *has to know already* how to create it (and thus how to stop global warming *now*). In other words, **if we don't know *exactly how* a certain technology will be created, we can't make the prediction that the technology *will be* created.**

Poincaré's Nonlinearities

Henry Poincaré, arguably the most highly regarded mathematician of the late 19th-century in France, contributed to the theory of unpredictability by proposing the existence of *nonlinearities*—small phenomena that, as time goes on, can have profound consequences.

A seminal illustration of the concept is the “three-body problem.” In a planetary system consisting of two planets (bodies), predictions about those planets' motion are easy to make: One need only account for the

relation between the planets. But if a *third* body enters the equation—perhaps a comet, passing between the two planets—the relation changes. Although the comet's effects may not be visible at first, the tiny forces it has exerted on the planets will compound with each cycle of the system, potentially resulting in massive changes.

Another example of nonlinearities is the so-called “butterfly effect.” In the 1960s, a meteorologist at MIT modeling weather patterns discovered that a minuscule rounding change in his input parameters resulted in tremendously different weather events. Thus the concept that **a butterfly flapping its wings in one part of the world can cause a hurricane in another**.

Each of these examples illustrates the difficulty of making accurate predictions in a “dynamical system,” wherein there are multiple variables and even slight changes can drastically alter results. **When it comes to dynamical systems, Poincaré noted that the properties of the system can be observed and discussed, but there's no way to compute—to use *numbers*—to describe the system.**

Suffice it to say, there is no more dynamical system than human society—Extremistan.

Hayek's Libertarianism

One of very few esteemed economists (he won a Nobel in 1974) to understand the futility of prediction, Fredrich Hayek spent much of his career railing against the main feature of socialism, central planning.

In a classical socialist society, all economic decisions—allocation of resources, setting of prices, etc.—are made by a single entity. **Hayek believed that a dynamical system like the economy was simply too complex for a single entity to master.**

Hayek attributed notions of central planning to misguided “experts” (*nerds*, in Taleb's terminology) who applied the methods of the physical sciences to social matters. Hayek recognized a stark distinction between soft sciences like economics and hard sciences like physics.

Hayek's skepticism regarding any entity's ability to predict the functioning of the economy led him to advocate for an “a-Platonic” approach, one that is open-minded and proceeds from the bottom up rather than the top down. He argued that **a libertarian system, wherein individuals are able to pursue their self-interest with a minimum of direction from above, is the best way to manage uncertainty**. In a system like this, he believed, the *system itself* would react accordingly to any unexpected changes in, say, the food or credit supply.

Hayek's anti-Platonifying approach to economics also puts his thought at odds with contemporary mainstream economists'.

For example, most economists today subscribe to some version of rational choice theory, which holds that individuals (or “agents”), when faced with an economic choice, will choose the option that maximizes their economic benefit. (A basic illustration: If an economic agent has a choice between an investment that returns \$2 and one that returns \$5, the agent will always choose the one that returns \$5.) The assumption that agents will always act rationally when it comes to economic choices allows economists to create models and propose predictions for future economic behavior.

The problem with rational choice theory, however, is that it's demonstrably *false*: **Empirical psychologists have shown that, in all sorts of economic situations, given various factors, individuals will act contradictorily and irrationally—they make the “wrong” choice.**

The upshot is that there is *no such thing* as a “general theory” of the economy, thus economic prediction

is impossible.

Forget Heisenberg

When discussing uncertainty, experts will frequently invoke physicist Werner Heisenberg's uncertainty principle. In its specific application, the uncertainty principle holds that the more precisely one measures the position of a given particle, the less precisely one can measure its momentum, and vice versa.

The problem with this analogy is that subatomic uncertainty actually hails from Mediocristan—particles' deviations from their expected positions or momentums can be graphed using a bell curve, and the sheer number and tininess of particles in existence mean those mild deviations average out.

Real uncertainty arises in the real world—like, for example, when Taleb can't reach his ancestral village in Lebanon from New York because Israel is battling Hezbollah.

Why Plan?

If prediction is impossible, and any human attempt to predict is ultimately futile, why do we continue to make forecasts or plans for the future?

First, it's important to note that **there are some areas of human experience where prediction is more viable than others**. For example, the physical sciences have an enviable track record in terms of predicting phenomena, from the subatomic to the cosmic.

But second, there is evidence that humans are *biologically conditioned* to anticipate. In fact, it's possible that our ability to imagine future consequences from present actions—an essential benefit of having *consciousness*—is the evolutionary advantage that allowed us to develop into the earth's most advanced species.

This advantage only serves us so far, however. In terms of avoiding immediate harm, our capacity for anticipation is essential. But when we try to anticipate the long-term workings of dynamical systems, we inevitably get ourselves into trouble.

Chapter 6: Predicting the Past

Through the limitations of inductive reasoning as illustrated by the turkey anecdote, as well as the distortions of the narrative fallacy and silent evidence, we've seen how problematic the past is vis-à-vis prediction. **But because of these phenomena and others, the past *itself* is as unknowable as the future.**

One of the major obstacles that prevents us from knowing the past with certainty is the impossibility of reverse engineering causes for events. That is, there's no way to determine the precise cause of an event when we work backward in time from the event itself.

An example should help illustrate.

Think of an ice cube sitting on a table. Imagine the shape of the puddle that ice cube will make as it melts.

Now think of a puddle on the table and try to imagine how that puddle got there.

The second thought experiment is much harder than the first. With the right physics know-how and ample time, one could model exactly what kind of puddle will result from the melting ice cube (based on the cube's shape, the environmental conditions, etc.). In contrast, it's nearly impossible to reverse engineer a cause from a random puddle (because the puddle could have been caused by any number of things).

When historians propose causes for certain historical events, they're looking at puddles and imagining ice cubes (or a spilled glass of water, or some other cause). The problem is that the sheer number of possible causes for a puddle—or a historical event—render any ascription of cause suspect.

Poincaré's nonlinearities too help illustrate this problem. Again, with the right tools and time, one might be able to observe how the flutter of a butterfly's wings in India causes a hurricane in Florida, **but it would be impossible to work backwards from the hurricane to that cause—there are just too many *other* tiny events that may have played a part.**

Our Information Is Always Incomplete

Mathematicians and philosophers draw a distinction between "true randomness" and "deterministic chaos." A "random" system is one whose operation is always and forever impossible to predict. A "chaotic" system is one whose operation *could* be predicted, but whose complexity makes prediction so difficult that it's effectively impossible. That is, if we had the right information, we would be able to make sense of "chaos."

Taleb notes that, for normal people trying to make predictions about their stock portfolio, for example, or the appreciation of the value of their house, there's no difference between "true randomness" and "deterministic chaos." **That's because when we're faced with a dynamical system, we *always* lack the necessary information to decide whether it's truly random or simply chaotic.**

Such is the case with history as well: Perhaps each major historical event conforms to some incredibly complex plan—that is, perhaps history is just chaotic, not random—**but we'll *never* have enough information to discern that plan.**

The Proper Use of History

Historical narratives are harmless if understood properly: as windows on to how human beings used to live and think rather than a source of actionable knowledge.

When we platonify history by treating it systematically, we let ourselves in for trouble. **We must be forever vigilant against (1) isolating causes for particular events and (2) using the past events and outcomes to guide our present and future choices.**

Chapter 7: What to Do When You Can't Predict

If we are surrounded by randomness and unpredictability, if our well-being is radically uncertain, what—besides despair—are our options?

1) Don't Sweat the Small Predictions

It bears repeating that humans' ability to predict in the short-term is unique among animal species and quite possibly the reason we've survived and thrived as long as we have. **To predict is human.**

So, when it comes to low-stakes, everyday predictions—about the weather, say, or the outcome of a baseball game—there's no harm in indulging our natural penchant for prediction: If we're wrong, the repercussions are minimal. It's when we make large-scale predictions and incur real risk on their basis that we get into trouble.

2) Maximize Possibilities for Positive Black Swans

Although the most memorable Black Swans are typically the negatively disruptive ones, **Black Swans can also be serendipitous.** (Shortform note: Love at first sight is undoubtedly a Black Swan.)

Taleb advocates (1) sociability and (2) proactiveness when presented with an opportunity as strategies for opening ourselves up to positive Black Swans. Sociability puts us in the company of others who may be in a position to help us—we never know where a casual conversation might lead. And proactiveness—taking up a successful acquaintance on an invitation to have coffee, for example—ensures we'll never miss our lucky break.

3) Adopt the "Barbell Strategy"

When Taleb was a trader, he pursued an idiosyncratic investment strategy to inoculate himself against a financial Black Swan. He devoted 85%–90% of his portfolio to extremely safe instruments (Treasury bills, for example) and made extremely risky bets—in venture-capital portfolios, for example—with the remaining 10%–15%. (Another variation on the strategy is to have a highly speculative portfolio but to insure yourself against losses greater than 15%.) The high-risk portion of Taleb's portfolio was highly diversified: **He wanted to place as many small bets as possible to increase the odds of a Black Swan paying off in his favor.**

The "barbell strategy" is designed to minimize the pain of a negative Black Swan while, potentially, reaping a positive Black Swan's benefits. If the market collapses, a person pursuing this strategy isn't hurt beneath the "floor" of the safe investments (say, 85%), but if the market explodes, he has a chance to capitalize by virtue of the speculative bets.

(Shortform note: There is, of course, the possibility that a Black Swan will affect even the safest investment—in fact, if we take Taleb at his word, *there is no such thing* as a safe investment.)

4) Distinguish Between Positive Contingencies and Negative Ones

Different areas of society have different exposure to Black Swans, both positive and negative.

For example, scientific research and moviemaking are "positive Black Swan areas"—catastrophes are rare, and there is always the possibility of smashing success. The stock market or catastrophe insurance, meanwhile, are "negative Black Swan areas"—upsides are relatively modest compared to the possibility of financial ruin.

It's important to note that the history of a negative-Black Swan area will underrepresent the possibility of catastrophe. An obvious example is financial markets, which always appear stable until they crash. Positive-Black Swan areas, oppositely, will underrepresent the possibility of serendipity: Despite years of research, a cure for cancer hasn't been found; but one *could* be found any day.

Suffice it to say, **we should take more risks in a positive Black Swan area than in a negative Black Swan one.**

5) Prepare, Don't Predict

Because Black Swans are, by definition, unpredictable, **we're better off *preparing* for the widest range of contingencies than *predicting* specific events.**

That's because, though Black Swans themselves can never be predicted, their *effects* can be. For example, no one can predict when an earthquake will strike, but one can know what its effects will be and prepare adequately to handle them.

The same goes for an economic recession. No one can predict precisely when one will occur, but, using the "barbell strategy" or some other means of mitigating risk, we can at least be prepared for one.

The Side-Effects of a Black Swan Obsession

Taleb's preoccupation with Black Swans—and his fellow humans' persistence in discounting them—has caused him to develop a contrary personality: where others see safety and stability, he sees risk and the potential for catastrophe; where others see volatility, he sees opportunity.

Taleb's guiding principle is simple: **He maximizes his exposure to positive Black Swans in all aspects of his life—socially, financially, aesthetically—and minimizes exposure to negative ones.**

In terms of investing, this means remaining highly skeptical of "safe" bets—blue-chip stocks, a perpetually rising stock market—and optimistic about speculative bets like venture capital. Because the speculative bets are transparently risky, Taleb knows to limit his exposure. The risks with "safe" investments, however, are hidden and therefore more worrisome.

Exercise: The Barbell Strategy

A barbell strategy devotes the majority of resources to safe options, and a minority to highly risky options that can pay off big. How can you integrate this into your life?

Write down a goal you've recently set for yourself, whether in your personal or professional life.

How do you plan to achieve that goal? Explore your current strategies for making that goal a reality.

Now, in the box below, explore how you might use Taleb's barbell strategy to achieve your goal. For example, if your goal is to secure a raise, you might devote 90% of your efforts to a safe bet (like improving your performance in your current role) and 10% to a riskier strategy (like taking on additional responsibilities).

Exercise: Maximizing Positive Black Swans

Think about how you can use randomness to your advantage.

Write down an area of your life where you think you could use improvement and why.

Brainstorm some positive Black Swans that could accelerate this improvement (remember that positive Black Swans are serendipitous moments that have a positive impact on your life). For example, if the area you chose was "romance," a positive Black Swan could be "finding a life partner."

Now brainstorm some ways you might increase the probability of these Black Swans occurring. For example, if your positive Black Swan was "finding a life partner," you might list "go to more parties" or "sign up for a class."

Appendix: The Contours of Extremistan

Although one needs only an intuitive sense of phenomena like wealth and market returns to understand that they don't adhere to the same rules as phenomena like height and weight, throughout the book Taleb provides a robust theoretical and statistical scaffolding for his claims about the differences between Mediocristan and Extremistan.

Because these discussions tend toward the technical and aren't essential for understanding Black Swans and their role in our lives, we at Shortform have decided to summarize them as an appendix.

Unfairness in Extremistan

As exemplified by figures like Beyoncé and Jeff Bezos, social and economic advantages accrue highly unequally in Extremistan.

One reason for this disparity is the "superstar effect." Coined by economist Sherwin Rosen to describe the unequal distributions of income and prestige in Extremistan sectors like stand-up comedy, classical music, and research scholarship, the "superstar effect" operates when marginal differences in talent yield massive rewards.

The superstar effect, it's vital to note, is meritocratic—that is, those with the most talent, even if they're only slightly more talented than their competitors, get the spoils. **What a theory like Rosen's fails to take into account, however, is that all-important aspect of life in Extremistan: *dumb luck*.**

Consider research scholarship for instance. Sociologist Robert K. Merton has observed that academics rarely read all the papers they cite; rather, they look at the bibliography of a paper they *have* read and pick sources to cite more or less at random.

Now imagine a scholar cites three authors at random. Then a second scholar cites those same three authors, because the first author cited them. Then a third does the same thing. Suddenly the three authors that have been cited are considered leaders in their field, all by dint of dumb luck.

The phenomenon identified by Merton's study has been called both "cumulative advantage" and "preferential attachment." These concepts describe **the innate human tendency to flock to past successes, regardless of whether those successes are the product of merit or chance.**

Although cumulative advantage/preferential attachment provides a better account of unfairness than the superstar effect—because it accounts for randomness in the distribution of advantage/preference—it still isn't a perfect theory of Extremistan unfairness. This is because, according to cumulative advantage/preferential attachment, winners stay on top. Once an entity—a company, an academic, a professional sports coach—reaches a certain level of success, cumulative advantage/preferential attachments holds that that entity will *continue* to be successful, because humans naturally favor past success. But this doesn't reflect reality.

For example, according to cumulative advantage/preferential attachment, Apple will forever be the king of consumer electronics, and Google will forever own the Internet. But even a cursory knowledge of business history shows a belief like this to be misguided. Consider this: If you tracked the 500 largest U.S. companies from 1957 to 1997, you'd discover that only 74 lasted the full 40 years. Some ceased to exist by virtue of mergers, but most simply failed. **In other words, even in Extremistan, giants can occasionally be toppled by dwarfs.**

Nevertheless, despite the possibility of creative destruction, Extremistan is defined by extreme *concentration*—of prestige, income, capital, influence, size, or what have you. **These concentrations create the appearance of stability—larger, more powerful entities are more resilient—but they actually create the potential for catastrophic black swans.**

The example par excellence of the risks of concentration is the banking industry. The explosive growth and interrelation of the major global financial institutions were the catalysts for the Great Recession of 2008.

(Shortform note: The popularity of *The Black Swan* is due, in large part, to Taleb's clairvoyance about the vulnerabilities of the financial industry. The book was published in April 2007, before the worst of the crisis, yet Taleb describes the contours of that crisis with eerie precision.)

The most far-reaching unfairness of Extremistan, however, more so than economic inequality, is inequality in intellectual influence. As illustrated by Merton's citation example just above, intellectual influence can be a matter of dumb luck rather than ability or insightfulness. This problem becomes acute when we trust thought leaders—"experts"—with our lives and livelihoods (see Chapter 4: The Scandal of Prediction).

The Limits of the Bell Curve

The classic bell curve—which is also called the "Gaussian distribution" after German mathematician Carl Friedrich Gauss—is an accurate description of Mediocristan phenomena, but it is dangerously misleading when it comes to Extremistan.

Consider human height, an eminently Mediocristan phenomenon. With every increase or decrease in height relative to the average, the odds of a person being that tall or short decline. For example, the odds that a person (man or woman) is three inches taller than the average is 1 in 6.3; 7 inches taller, 1 in 44; 11 inches taller, 1 in 740; 14 inches taller, 1 in 32,000.

It's important to note that the odds not only decline as the height number gets further and further away from the average, but they decline at an accelerating rate. For example, the odds of someone being 7'1" are 1 in 3.5 million, but the odds of someone being just four inches taller are 1 in 1 billion, and the odds of someone being four inches taller than *that* are 1 in 780 billion!

Now consider an Extremistan phenomenon like wealth. In Europe, the probability that someone has a net worth higher than 1 million euros is 1 in 62.5; higher than 2 million euros, 1 in 250; higher than 4 million, 1 in 1,000; higher than 8 million, 1 in 4,000; and higher than 16 million, 1 in 16,000. **The odds decrease at a constant, rather than accelerating, rate, indicating that their distribution doesn't conform to a bell curve.**

(Note: The European wealth statistics cited above aren't precise, but they illustrate the central point: that wealth is scalable and does not look like a bell curve.)

The upshot is that **in Extremistan, as the name suggests, extreme events have much better odds of occurring than in Mediocristan. Simply put, the bell curve does not apply.**

The Mistakes of Adolphe Quételet

Adolphe Quételet (1796–1874) was a French mathematician who developed the idea of the "average human" (*l'homme moyen*) through the use of "means"—golden averages that represented the ideal human form.

At first, his inquiries were limited to human beings' physical characteristics—height, weight, newborn weight, chest size—but before long, he began to seek averages in the social realm by studying human habits and morals. Quételet developed bell curves to describe humans' deviance—literally—from both physical and moral norms.

Although Quételet's ideas were influential in his historical moment, there were skeptics. Augustin Cournot, a fellow mathematician and philosopher, pointed out that social averages would necessarily depend on exactly *whom* and *what attribute* was being studied. An obvious example is human diet: If you were trying to determine the amount of seafood an average human consumes, you might get vastly different measurements if you were surveying inland populations versus coastal ones, or populations whose cultural traditions included a seafood-heavy diet.

In our contemporary moment, the most egregious misuse of bell curves can be found among economists and financial-industry analysts. Most economists, including many Nobel Prize-winners, treat financial markets as though they hail from Mediocristan when, in fact, they're from Extremistan.

A stunning example of economists' misinterpretation of the market came in 1998, with the collapse of Long-Term Capital Management (LTCM), a hedge fund that included among its founding partners two Nobel Prize-winners, Myron Scholes and Robert C. Merton (Robert K. Merton's son). Scholes and Merton won their Nobel Prize for a theory of stock option pricing; the theory, however, was based on a Gaussian model of the market that excluded the possibility of Black Swans.

When markets were disrupted by the Asian financial crisis of 1997 and the Russian financial crisis of 1998—a one-two punch of Black Swans—Scholes and Merton's theory was revealed for the fantasy it was. The twin crises resulted in a crushing \$4.6 billion loss for LTCM in fewer than four months. The firm had to be bailed out by a consortium of private banks, at the behest of the Federal Reserve.

The problem is that economists' models are thoroughly Platonified: self-consistent, but bearing almost no resemblance to reality. **Economists are like philosopher John Locke's madmen: they reason "correctly from erroneous premises."**

Representing Randomness: Fractals and Power Laws

So can the phenomena of Extremistan be modeled—and thereby predicted—at all? Not precisely; but they *can* be approximated—by the use of *fractals*.

Developed by mathematician Benoît Mandelbrot in the 1970s, "fractals" are Mandelbrot's coinage for geometric patterns that repeat at different scales. Fractals, unlike pure geometric shapes like triangles or circles, are seen quite frequently in nature. For example, a leaf's veins look like little branches, and a tree's branches look like little trees: the basic shape of the tree is echoed at the smaller scales.

The reason fractals are helpful in representing Black Swans is that their internal ratios stay constant across scales. Unlike bell curves, in which ratios decline at accelerating rates the further one gets from the average, fractals exhibit no (or mild) acceleration. They obey *power laws*, which describe a functional relationship between two quantities in which the relationship remains constant no matter what the initial size of the quantities.

Take the European wealth example noted just above. As the wealth number doubled, the incidence decreased by 4x (wealth greater than 1 million: 1 in 62.5; wealth greater than 2 million: 1 in 250; wealth greater than 4 million: 1 in 1,000; etc.). The "power" in this relationship is 2, because 2^2 (2 squared) is 4.

Now, imagine the power was 1: Each time the wealth doubled, the incidence would only decrease by 2x ($2^1=2$). **The probability of extraordinary wealth would increase.**

For Extremistan phenomena, the power laws aren't known with any certainty, but they can be approximated. Say, for example, you want to assess the risk of a stock portfolio, and you know that, based on past data, the worst-case scenario is a -5% move once every 2 years. With a power of 2—remember that the power is estimated—you can assume that a -10% move will happen once every 8 years ($2^2=4$) and a -20 percent move will happen once every 32 years. (For scale, in the 1987 crash, the U.S. stock market lost almost 23% of its value, an *impossibility* according to Gaussian economic models.) **With fractal/power-law distributions, a 1000-year flood can become a 100-year one, and a 100-year one can become a 10-year one.**

When we begin to analyze Extremistan in terms of the fractal paradigm, some Black Swans suddenly become “gray”—they're not predictable with any sort of precision, but they are, at least, *imaginable*.

Exercise: Reflecting on The Black Swan

Reflect on your takeaways from Taleb's book.

Which of Taleb's concepts or examples did you find most *surprising* and why?

Which of Taleb's concepts did you find most *useful* and why?

Were there any concepts or claims that seemed spurious or incorrect to you? If so, explain where Taleb went wrong.